Reward scaling Due to the varying value ranges of rewards in different game environments/tasks, enhancing stability across a series of sequentially learned tasks is crucial in all environments. To address this, a single modification is made to the reward structure of the games, but only during the training phase. Given the substantial variability in the scale of scores from game to game, all positive rewards are set to be 1, and all negative rewards to -1, with 0 rewards remaining unchanged. This normalization facilitates more stable and consistent learning across varying tasks and game environments by mitigating the impact of extreme reward values, as demonstrated in [35].

Observations At each timestep, an agent only receives normalized visual input, which is represented as a 84×84 grayscaled image. The environment is configured for frame skipping, allowing the agent to interact less frequently with the environment. Observations from four consecutive frames are stacked together to form an observation with enhanced temporal dimensionality. Consequently, the agent's policy receives input in the format $[N, 4, 84, 84]$, where N represents the number of environments being used in parallel for training the agent in the progress phase with the A2C algorithm. Our implementation of A2C is based on PyTorch [44, 45] and incorporates concepts from [46, 47]. We used Bayesian Hyperparameter Optimization over selected configurations to maximize normalized scores across tasks. Our experiments indicate that TAPD demonstrates strong stability, with minimal performance variation across different hyperparameter settings and consistent behavior across configurations. Details on the hyperparameters of the learning architecture, algorithms, and experimental settings are provided in Appendix B.

5.2 The Intrinsic Reward in the Task-Agnostic Phase

We evaluate the performance of the agents based on task rewards, which are specifically designed to measure active interactions. Although extrinsic rewards are recorded, they are not used during the task-agnostic training phase. Instead, the agent focuses exclusively on optimizing the intrinsic rewards it generates. The task-agnostic phase receives tasks from the Meta-Environment, which includes BeamRider and SpaceInvaders. These tasks are uniformly sampled from the Meta Environment without any task boundaries [14]. We employ TAPD's task-agnostic phase, as it is the most general of all the three variations.

Fig. 3 illustrates the learning progress of TAPD during the task-agnostic phase. In this phase, the task-agnostic policy is compressed after a specific number of timesteps into the knowledge base. As shown in the figure, the agent's performance improves even without extrinsic rewards. Every 300k timesteps, the task-agnostic policy is distilled into the knowledge base, resulting in an exploratory repertoire. This exploratory knowledge is then utilized within the task-agnostic phase, using generalized knowledge to further generalize knowledge. When a task change occurs, the active column in the task-agnostic phase quickly adapts to the new environment, as indicated by the spikes in the graph. This suggests that the knowledge base is aware of the environment it is in, having learned the dynamics of the environment through the forward model that produces the intrinsic rewards that guide the action selection of the active column.

These rewards aid in further exploration of the environment, leading to the discovery of more novel states. The rapid task adaptation also highlights the mechanism of the knowledge base, as the active column remembers the previous situation without external goals from the environment with the help of the knowledge base. After the maximum number of timesteps in the task-agnostic phase has been reached, the alternation between progress and compress phases begins, but now with a pre-explored knowledge base, which accelerates the learning process for downstream tasks.

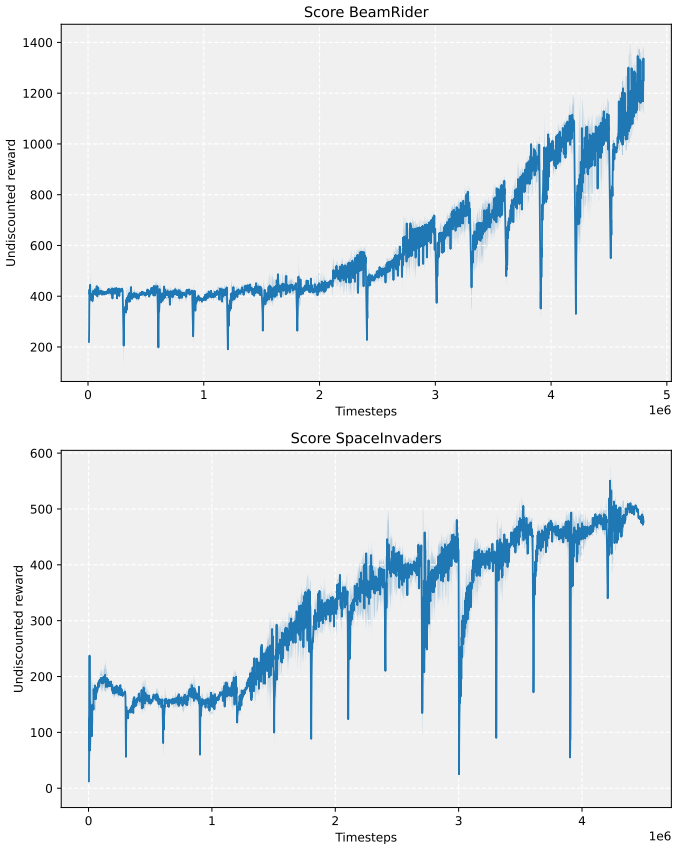

Fig. 3: Performance evaluation in the task-agnostic phase. The environment is uniformly sampled, indicating no task-boundaries. Runs averaged over 8 random seeds. Timesteps=300000 between distillation rounds in the task-agnostic phases. Averages are taken over 100 episodes.

5.3 Evaluation of Knowledge Accumulation and Transfer to Downstream Tasks

We compare our Task-Agnostic Policy Distillation (TAPD) algorithm with its task-agnostic phase, Online EWC [11, 13], Progressive Nets [48], and the Progress &

Compress baseline from [13]. This comparison aims to evaluate whether the task-agnostic phase can accelerate transfer learning in the progress phase for downstream tasks, while also improving sample efficiency. The task-agnostic phase, as depicted in Fig. 2, serves as the pre-training phase. The algorithms share the same architecture and hyperparameters, with the exception that TAPD has specific parameters for the task-agnostic phase. The tasks Pong, SpaceInvaders, BeamRider, DemonAttack, and AirRaid are sequentially learned. The evaluation focuses on the learning performance in the progress phase, as the emphasis lies on evaluating the positive forward transfer and sample efficiency of the active column. In the task-agnostic phase, TAPD uniformly samples SpaceInvaders and BeamRider, simulating tasks with undefined boundaries Zeno et al. [14]. On each visit to a task, the active columns parameters and lateral connections are re-initialized.

Fig. 4 shows the learning curves of Pong, SpaceInvaders, BeamRider, DemonAttack, and AirRaid along with their corresponding entropies. In the case of Pong, although TAPD hadn't previously learned the game, its knowledge base enabled much faster exploration of newly encountered tasks, resulting in significantly higher performance than all other algorithms, which showed only slight improvement after 1.5 million timesteps. On the other hand, TAPD already learned to adapt to the tasks at around 0.6 million timesteps, showcasing the high sample efficiency of the proposed approach. This adaptation is also evident in the entropy of the distribution for solving the Pong task. The entropy initially starts lower than the Progress & Compress baseline, indicating that the agent is aware of the required movements. However, the Progress & Compress baseline exhibits a similar pattern throughout the training process, albeit with lower performance compared to TAPD. In later timesteps, TAPD exhibits a more exploratory behavior compared to the Progress & Compress baseline, with occasional decreases. The knowledge base of TAPD seems to retain its exploratory behavior during the learning process of Pong. The entropy in Progressive Nets exhibits a steep learning curve during the second visit, which is expected. This is because the dedicated column used to train the Pong task in the first visit retains and continues to refine the parameters associated with Pong during the second visit, leading to rapid progress, which extends until the third and final visit. The Online EWC algorithm demonstrates a recurring pattern in the initial timesteps of each visit, where the score increases, indicating that the model detects the task but is unable to further improve upon it.

Moving on to the SpaceInvaders task, TAPD appears to remember this environment, which potentially contributes to its superior performance compared to all other algorithms in the initial phase. Throughout the training process, TAPD maintained lower entropy compared to the Progress & Compress baseline, which showed much higher entropy across all three visits. Despite the Progress & Compress baseline employing lateral connections in subsequent visits, its performance does not improve over time due to the high entropy of its policy, which suggests ongoing exploration. This does not mean the baseline is stuck, it may improve its performance after many more timesteps, but this would indicate lower sample efficiency. A similar behavior is observed with the Online EWC algorithm. The Progressive Nets algorithm demonstrates a strong initial performance with no interference from task-specific learned